LTI Systems

$x(n) = \cos(\omega_0 n)$ → [LTI, $\{h(n)\}$ (Real Value)] → $y(n) \overset{?}{=} A\cos(\omega_0 n + \theta)$

($n \in Z$)

Same as input ↓ Phase shift ↓

↓ Amplitude gain

$\{h(n)\} \overset{DTFT}{\longleftrightarrow} H(\omega)$

imp. resp — freq. resp

Recall eigen function

$e^{j\omega_0 n} \xrightarrow{LTI} H(\omega_0)\, e^{j\omega_0 n}$

$H(\omega_0)$ → frequency response of system at $\omega = \omega_0$

$$\cos(\omega_0 n) = \frac{e^{j\omega_0 n}}{2} + \frac{1}{2} e^{-j\omega_0 n}$$

LTI ↓ → $H(\omega_0)\frac{e^{j\omega_0 n}}{2}$

LTI ↓ → $H(-\omega_0)\frac{e^{-j\omega_0 n}}{2}$

Since $\{h(n)\}$ is real valued,

$H(\omega)$ is conjugate symmetric

$H(-\omega) = H^*(\omega)$

$$H(\omega_0) = A e^{j\theta}$$

Output $y(n) = \frac{1}{2} A e^{j\theta} e^{j\omega_0 n} + \underbrace{\frac{1}{2} A e^{-j\theta} e^{-j\omega_0 n}}_{H^*(\omega_0)}$

$= A \cos(\omega_0 n + \theta)$

x ——————————— x

causal inputs

$x(n) = \cos(\omega_0 n)\, u(n)$

$= \frac{e^{j\omega_0 n}}{2} u(n) + \frac{e^{-j\omega_0 n}}{2} u(n)$

$a^n u(n)$ $\quad$ $b^n u(n)$

Generic Input

(causal) $x(n) = \sum_{k=1}^{Q} a_k^n u(n)$

$a_k \neq b_m$

Impulse resp (Causal) $h(n) = \sum_{m=1}^{P} b_m^n u(n)$

⇓

rational Transfer function $H(z)$ ⊛ (deg. of Nr is smaller than degree Dr)

Z - Transform of output

$$Y(z) = X(z) \cdot H(z)$$

= rational (degree of Nr < degree of Dr)

$$= \left(\sum_k \frac{1}{(1 - a_k z^{-1})} \right) \left(\sum_m \frac{1}{(1 - b_m z^{-1})} \right)$$

(Partial fraction)

$a_k \Rightarrow$ poles of input

b_m poles of system

$$y(n) = \sum_{k=1}^{Q} C_k \, a_k^n u(n) + \sum_{m=1}^{P} D_m \, b_m^n u(n)$$

input modes — system modes

For stable system:

$|b_m| < 1$ for all m

System modes converge to zero (transient mode) for large n.

If $|a_k| = 1$, input modes will reach steady state (will not go to zero)

If input mode = system mode

(resonance)

$$x(n) = a^n u(n) \qquad h(n) = a^n u(n)$$

$$H(z) = X(z) = \frac{1}{1 - az^{-1}} \; ; \; |z| > |a|$$

$$Y(z) = \frac{1}{(1 - az^{-1})^2} \; ; \; |z| > |a|$$

$$y(n) = (n+1)\, a^n u(n)$$

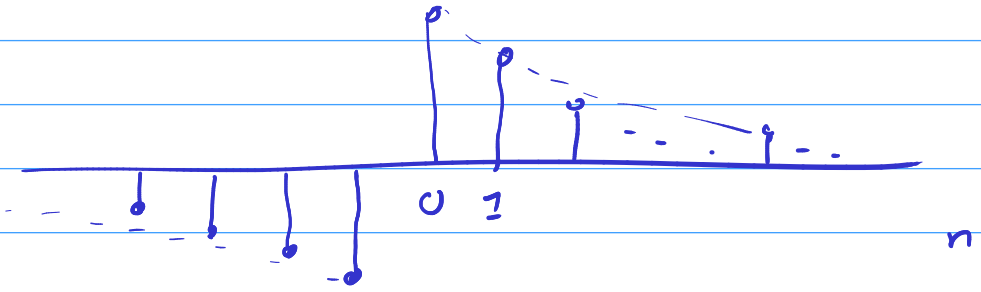
$\therefore$ resonance.



Unit step Signal

DTFT of $u(n)$

Consider $x(n) = a^n u(n) - \left(\frac{1}{a}\right)^n u(-n-1)$ $\quad 0 < a < 1$



$$X(z) = \frac{1}{1-az^{-1}} + \frac{1}{1-(1/a)z^{-1}}$$

$|z| > a \qquad |z| < 1/a$

ROC : $a < |z| < 1/a$

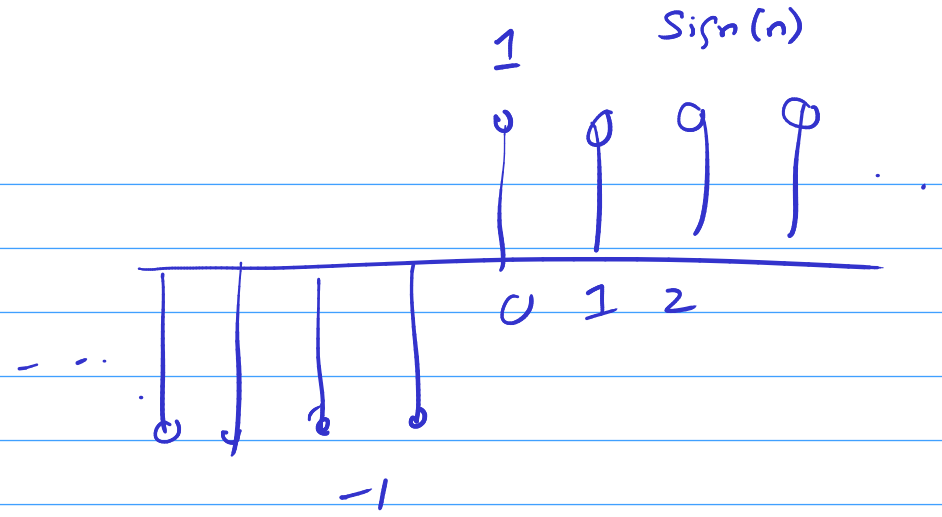
ROC includes unit circle.

DTFT $\quad X(\omega) = \frac{1}{1-ae^{-j\omega}} + \frac{1}{1-\frac{1}{a}e^{-j\omega}}$

Take $\lim_{a \to 1}$ $\quad x(n) = \begin{cases} 1 ; & n \geq 0 \\ -1 ; & n < 0 \end{cases}$

$= sign(n)$

$$X(\omega) = \frac{2}{1-e^{-j\omega}}$$



$$u(n) = \frac{1 + Sign(n)}{2}$$

$$= \frac{1}{2} + \frac{Sign(n)}{2}$$

(DC freq)

$$U(\omega) = \pi\delta(\omega) + \frac{1}{1 - e^{-j\omega}} \qquad -\pi < \omega < \pi$$

$\pi\delta(\omega)$ → dirac delta.

Evaluating integrals

$$x(n) \xleftrightarrow{DTFT} X(\omega)$$

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(\omega)\, e^{-j\omega n}\, d\omega = x(n) \qquad n = 0$$

$$x(0)$$

$$1$$